Chem-330 Tutorial-2 Total marks 30, Time 1hr 20 min

1. (a) Reaction between methyl bromide with aq. NaOH gives methanol. How do you prove that the reaction follows S_N2 path showing an evidence?

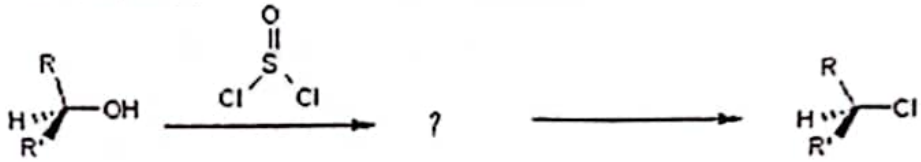
(b) The following compound is not hydrolyzed even by a hot and aqueous solution of KOH.



(c) Why does S_N2 reaction rate decrease as the following series is traversed?

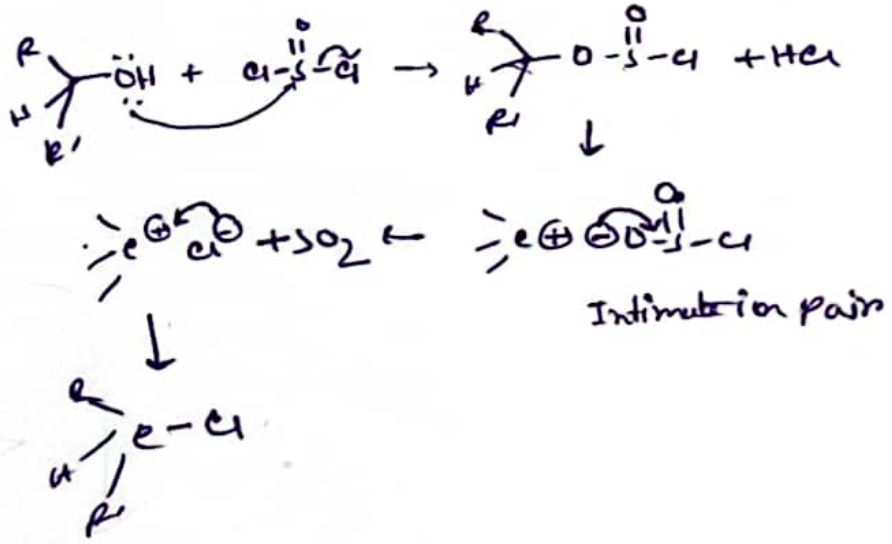
CH_3-Cl $MeCH_2$-Cl Me_2CH-Cl Me_3C-Cl

(d) Complete the following reaction showing the intermediate. Explain the mode of reaction that properly justify the stereochemical aspect (retention) of the product.



(e) The reaction of neopentyl bromide with aqueous NaOH gives 2-methyl-2-butanol. Give one evidence in support the rearrangement takes place at time of the reaction.

2. Bayer-Villiger oxidation is very useful in different synthetic fields. Write your ansewer on it mentioning statement, mechanism, application etc.



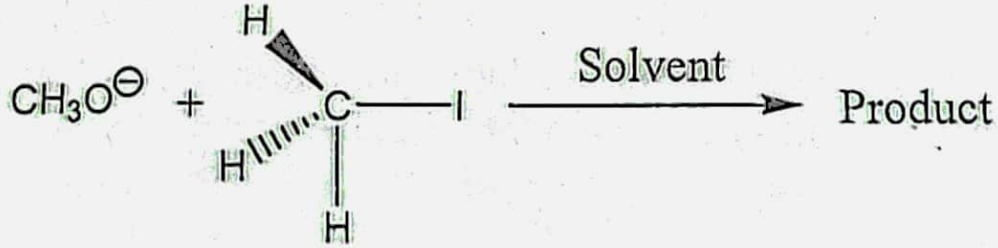
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Sample

* (a) What is the salient difference between S_N1 and S_N2 pathway?
(b) How do you make a kinetic distinction between the operation of the S_N1 and S_N2 modes.
(c) The rate of solovolysis of the tertiary halide, Me3CBr, is found to be 3×10^4 times faster in 50% aqueous ethanol than in ethanol-explain.
(d) For the S_N2 mode, increasing solvent polarity is found to result in slight decrease in reaction rate-why?
(e) Why does polar non hydroxylic DMF have marked effect on (increased 4.5×10^4) S_N2 reaction of MeI with N_3^- at 0° C compared the reaction with MeOH?
(f) Why does S_N2 reaction rate decrease as the following series is traversed?

CH_3-Br $MeCH_2$-Br Me_2CH-Br Me_3C-Br

3. Complete the reaction and answer the following questions: 1.0

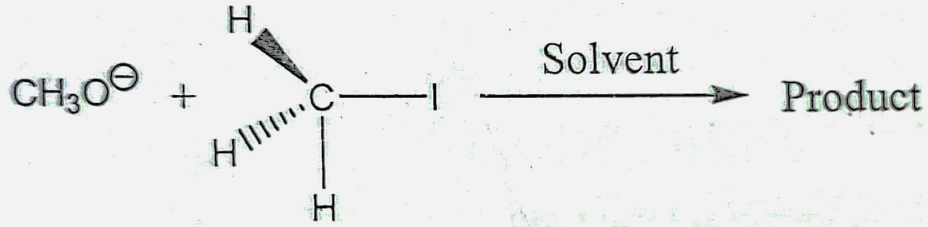


(i) Does the reaction follow the S_N2 mechanism? If so, then give evidence in support of your answer. 2.0

(ii) Propose a mechanism for the reaction? 2.0

(iii) If the above reaction follows the S_N2 mechanism and the iodine atom of the above methyl iodide is bonded to a chiral carbon atom then how many stereoisomeric products will be formed? Explain with suitable example. 3.0

3. Complete the reaction and answer the following questions: 1.0



(i) Does the reaction follow the S_N2 mechanism? If so, then give evidence in support of your answer. 2.0

(ii) Propose a mechanism for the reaction? 2.0

(iii) If the above reaction follows the S_N2 mechanism and the iodine atom of the above methyl iodide is bonded to a chiral carbon atom then how many stereoisomeric products will be formed? Explain with suitable example. 3.0

(iv) If the above reaction is independently carried out in the presence of DMSO and H_2O then what will be the effect of these solvents on the rate of the reaction? Give explanation in support of your answer. 3.0

(v) If CH_3OH is used as a nucleophile in the above reaction instead of methoxide ion (CH_3O^-) and the reaction is independently carried out in the presence of DMSO and H_2O then what will be the effect of these solvents on the rate of the reaction? Give explanation in support of your answer. 3.0

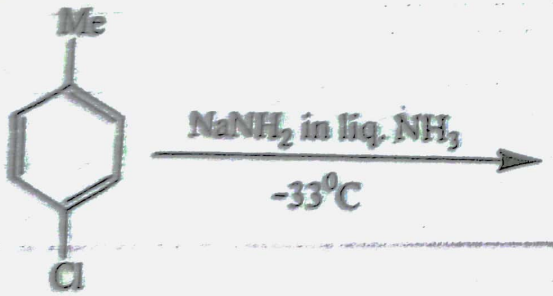
5. (a) What are the salient differences between S_N1 and S_N2 reactions?

(b) The S_N2 hydrolysis of neopentyl bromide is slow but under S_N1 mode the rate is increased and found an unexpected product, 2-methyl-2-butanol - justify the statement.

(c) How can you prove that S_N2 reactions are attended by inversion of configuration?

(d) In substitution reaction, neighbouring group participation leads to retention of configuration - explain with suitable example.

(e) Complete the following reaction and give mechanism with evidence.

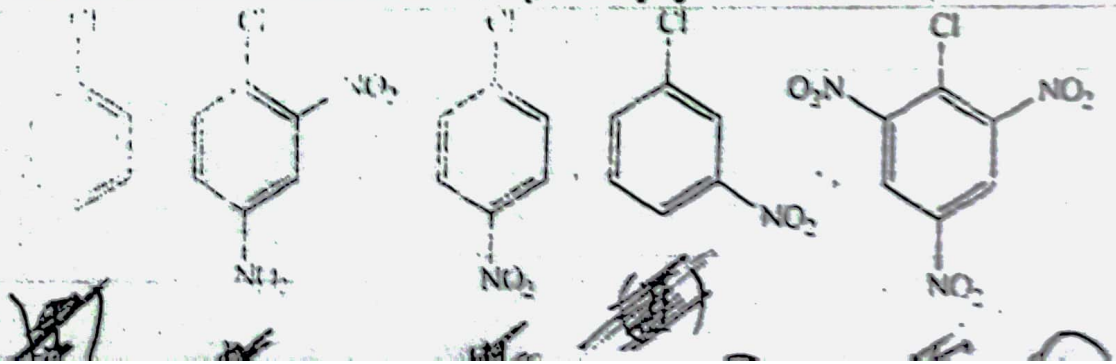


2nd Tutorial Examination - 2018

Course No.: 330, Full Marks: 20, Time: 60 minutes

Date: 12 December, 2018

1. What are the salient differences between S_N1 and S_N2 reactions? 3
2. The rate of solvolysis of the tertiary halide, Me_3CBr, is found to be 3×10^4 times faster in 50% aq. ethanol than in ethanol alone - explain. 2
3. The S_N2 hydrolysis of neopentyl bromide is slow but under S_N1 mode the rate is increased and found an unexpected product, 2-methyl-2-butanol – justify the statement. 3
4. How can you prove that S_N2 reactions are attended by inversion of configuration? 3
5. In substitution reaction, neighbouring group participation leads to retention of configuration - explain with suitable example. 3
6. Nucleophilicity is found to follow the sequence: $I^- > Br^- > Cl^-$. How can you explain it? 2
7. Arrange the following compounds in order to increase the rate of nucleophilic aromatic substitution reaction and justify your answer. 4



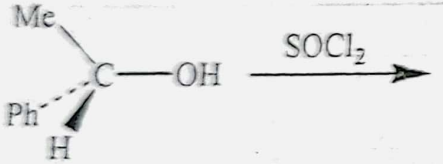
5. (a) Which reagent in each of the following pairs is more nucleophilic and why? 2

(i) $(CH_3)_3B$ and $(CH_3)_3N$ (ii) H_2O and H_2S

(b) Give the reaction mechanism of the hydrolysis of *t*-butyl bromide in presence of base. 2

(c) Solvolysis of (+)-PhCHMeCl leads to 98% racemization in aqueous acetone while (+)-C_6H_{13}CHMeCl leads to only 34% racemization. Explain. 3

(d) What is S_Ni? Complete the following reaction and show its mechanism. 3



(e) Compared to allyl chloride, vinyl chloride is very unreactive towards nucleophilic substitution reaction - explain. 3

(f) Benzyne is very unstable and reactive. Why? 1

7. (a) What is the salient difference between S_N1 and S_N2 pathway? 3

(b) How do you make a kinetic distinction between the operation of the S_N1 and S_N2 modes. 1

(c) The rate of solovolysis of the tertiary halide, Me_3CBr, is found to be 3×10^4 times faster in 50% aqueous ethanol than in ethanol-explain. 2

(d) For the S_N2 mode, increasing solvent polarity is found to result in slight decrease in reaction rate-why? 2

(e) Why does polar non hydroxylic DMF have marked effect on (increased 4.5×10^4) S_N2 reaction of MeI with N_3^- at 0° C compared the reaction with MeOH? 3

(f) Why does S_N2 reaction rate decrease as the following series is traversed? 3

CH_3-Br $MeCH_2$-Br Me_2CH-Br Me_3C-Br

3.5 ×4=14